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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE403/20EE403 – MICROCONTROLLER

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the architecture of SST89x516RD2 with relevant block diagram.	8	L*2	1	1,5
b)	With example explain addressing modes used in embedded C programming.	6	L2	1	1,5
c)	Write a C program to generate a 5kHz square wave on P2.1 with 25% duty cycle. Use Timer1 in mode2. Assume crystal frequency of 11.0592MHz.	6	L3	2	1,5
2. a)	Mention any four technical specification of SST89x516RD2 controller.	4	L1	1	1,5
b)	Write a C program to configure timer 0 in mode 1 and generate a 100Hz square wave in port pin P2.4. Assume Fosc = 12MHz.	8	L3	2	1,5
c)	Mention the various bits of TCON and TMOD register and also explain the significant of each bits.	8	L2	2	1,5
3. a)	With neat block diagram explain the configuration of timer in mode 0 and also write different steps to configure the timer in mode 0.	8	L2	2	1,5
b)	Elaborate the port structure of PORT0 with diagram.	6	L2	1	1,5
c)	Write a program to generate 2 square waves with one of 5kHz in P1.2 and another of 25kHz on P2.3. Use Timer in mode2. Assume f= 12MHz. Use interrupt.	6	L3	2	1,5
	Unit – II				
4. a)	Write the various steps required to transfer the data serially	4	L1	3	1,2,3,5
b)	Write a C program to send a message 'ELECTRICAL' serially. Use serial mode 1, baud rate 9600 and assume crystal frequency 11.0592MHz.	6	L3	4	1,2,3,5
c)	Write a program to display digits key value in both seven segment displays when the corresponding key in hex keypad is pressed	10	L3	4	1,2,3,5
5. a)	Explain the significance SBUF and SCON register.	4	L2	3	1,2,3,5
b)	Write a program to display "NITTE" in line 1 of 16x2 LCD connected to P0. The control pins RS and E of LCD are connected to P3.6 and P3.7 respectively.	10	L3	4	1,2,3,5
c)	Write a program to receive an array of data NMAMIT serially with 2400 baud rate. Assume f=11.0592MHz	6	L3	3	1,2,3,5
6. a)	Explain the importance of TI, RI flags.	4	L2	3	1,2,3,5
b)	Write a program to display up counter continuously in seven segment display from 00 to FF with a delay of 1 second.	8	L3	4	1,2,3,5

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- c) Write a program to display analog voltage of potentiometer connected to channel 0 of ADC in LED array connected to Port 1 of microcontroller. The ADC data lines are connected to P0. The control lines OE, Start and EOC are connected to P2.5, P2.6 and P2.7 respectively.

8 L3 4 1,2,3,5

Unit – III

7. a) With detailed block diagram and circuit diagram, design Microcontroller based automated irrigation system.
b) With detailed block diagram and circuit diagram, design a 8051 based system to measure RMS value of AC voltage upto 250V.
8. a) With detailed block diagram and circuit diagram, using microcontroller design a password based security system to restrict unauthorized person from entering to the firm.
b) With detailed block diagram and circuit diagram design a microcontroller based system to monitor speed of a motor and display it on LCD .if speed exceeds limit 500 RPM switch off the motor.

10 L4 5 1,2,3,5

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
